

AYMANE AIT DADS

Machine Learning Systems Intern | LLM Fine-Tuning · Computer Vision · AI Research

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by fine-tuning a 30B-parameter Mamba-Transformer with LoRA SFT on structured multi-category reasoning tasks - a result achieved while the competition remains ongoing through June 2026. Fine-tuned a CLIP ViT-L/14 backbone (428M parameters) for AI-generated image detection, ranking 1st in the EURECOM class leaderboard and submitting to the NTIRE 2026 benchmark at CVPR, achieving 0.791 AUC across 25+ generator types. Combines deep technical fundamentals in LLM fine-tuning, computer vision, and ablation methodology with a proven willingness to challenge assumptions - directly matching Meshy's expectation of innovation-driven ML engineers. Aims to apply reasoning-model research and generative AI expertise to Meshy's 3D content generation pipeline, advancing world-class AI-graphics product development.

CORE COMPETENCIES

Machine Learning Systems | Technical Fundamentals | AI Research & Innovation | LLM Fine-Tuning | Computer Vision | Ablation Methodology | Generative AI | Full-Time Commitment

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built **Random Forest + K-Means machine learning models** on 100K+ fiber-optic network records to classify network quality across thousands of nodes with production-ready output.
- Developed an end-to-end clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, adopted directly by the network optimization team for operational decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation scripts, and reproducible Pandas/Scikit-learn pipelines across the full data workflow.
- Delivered stakeholder presentations translating **model outputs into maintenance recommendations**, communicating technical findings clearly to non-technical business teams using Power BI dashboards.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) across 5 structured reasoning categories.
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built a diagnostic evaluation pipeline revealing that reasoning over-expansion - not incorrect reasoning - was the primary failure mode in hard tasks.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in the EURECOM class private leaderboard** (0.791 AUC) and submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark by fine-tuning a CLIP ViT-L/14 backbone (428M parameters) with a custom MLP head and dual learning-rate strategy across 250K images and 25+ generator types.
- Discovered that **1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC)** for out-of-distribution generalization; applied 10-view TTA ensemble weighting to maximize robustness across unseen generative model architectures.

PyTorch · OpenCLIP · CLIP ViT-L/14 · AdamW · FP16 Mixed Precision

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Image Security, Cloud Computing, Statistics, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & Reasoning: LoRA / QLoRA | SFTTrainer | PEFT | Unsloth | vLLM | chain-of-thought | Mamba-Transformer | completion-only loss | BF16 precision

ML / Computer Vision / NLP: PyTorch | TensorFlow | Scikit-learn | CLIP ViT | Hugging Face Transformers | OpenCLIP | CNN / DenseNet | TRL | Triton

Data Science & MLOps: Python | Pandas | NumPy | ablation studies | feature engineering | ensemble methods | Git | Linux | CUDA | Power BI